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ORIGINAL RESEARCH

Microscopic and Biological Characterization of Cryopreserved Equine Umbilical Tissue Allograft

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Purpose: Equicenta®CTM is a cryopreserved micronized equine umbilical-derived tissue allograft (eUTA) that is minimally processed and contains an abundance of fetal proteins in a connective tissue matrix. This study aimed to characterize this eUTA's microscopic structure, biodegradation, and potential to release biologically relevant proteins from a tissue scaffold reservoir, including cytokines, growth factors, and extracellular vesicles (EVs).

Methods: Microscopic evaluation of eUTA was performed using stereoscopic culture and pelleted smears. In vitro enzymatic digestion with collagenase or trypsin (positive control) in tissue culture suspensions was used to assess morphological changes. A 24-hour digestion study quantified the total protein, collagen type 1 alpha chain 1 (Col1A1), and tissue inhibitor of metalloproteinase-2 (TIMP-2). Anabolic, anti-inflammatory, and anti-catabolic biomediators were measured using multiplex ELISA before and after detergent extraction. EVs were evaluated by nanoparticle tracking analysis and tetraspanin detection.

Results: Microscopy revealed free-floating microparticle aggregates and fibrillar net-like extracellular matrix scaffolds in a proteinaceous ground substance. Collagenase caused gradual matrix breakdown, whereas trypsin resulted in rapid, complete degradation. The eUTA contained multiple cytokines and growth factors; selected anabolic and anti-catabolic proteins increased after scaffold digestion or extraction, including insulin-like growth factor (IGF-1), alpha-2 macroglobulin (A2M), and TIMP-2. (Col1A1) was released continuously over 24 hours during collagenase digestion. EV analysis confirmed abundant particles expressing CD9, CD63, and CD81.

Conclusion: Microscopic evaluation of eUTA revealed an intricate filamentous microparticle matrix capable of serving as a scaffold for endogenous tissue repair in orthopedic injuries. This matrix can be degraded by naturally occurring proteolytic collagenases, thereby releasing matrix proteins and bioactive factors over time, which may support healing.

Keywords: horse, orthobiologic, umbilical tissue, extracellular vesicles, cytokines, growth factors

Introduction

Equine musculoskeletal injuries of joints, tendons, and ligaments are a leading cause of lost training days, diminished performance, and early retirement,[1–4] and classical therapies often provide only symptomatic relief without reliably restoring normal structure and function. Orthobiological strategies that combine a rich repertoire of pro-reparative proteins with a biocompatible scaffold[5–7] are therefore of particular interest to support tissue healing while minimizing the adverse effects associated with repeated corticosteroid or non-steroidal anti-inflammatory drug administration.[8–11]

Microparticulate bio-scaffolds, such as umbilical tissue allografts, are increasingly recognized as dual-purpose biologics that can deliver a complex repertoire of bioactive proteins and provide structural support for endogenous tissue repair after injury.[12–14] Recent work in horses has demonstrated that this cryopreserved equine umbilical cord tissue allograft (eUTA) is highly consistent across donors.[15] The eUTA protein suspension has been characterized in detail, providing a defined backdrop for further mechanistic studies. Proteomic analysis of eUTA and decellularized Wharton's jelly from human umbilical cords has identified highly abundant proteins predominantly involved in structural support and protection of connective tissue, including multiple fibrillar collagens and matrix glycoproteins, along with

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moderately abundant collagens, glycan and proteoglycan components, and proteins linked to tissue repair, oxidative protection, and energy homeostasis.[15,16] Lower-abundance but uniformly expressed mediators, such as insulin-like growth factor-2 and clusterin, further suggest the capacity to influence matrix regulation and cell survival. Complementary immunoassays showed low levels of inflammatory cytokines and the presence of key trophic mediators, including chemokines, angiogenic factors, and hyaluronic acid, together with an anti-inflammatory profile characterized by interleukin-1 receptor antagonist (IL-1Ra) and interleukin-10 (IL-10).[15] These findings are similar to reports evaluating human amnion/chorion and umbilical tissue allograft, where various growth factors, such as basic fibroblastic growth factor (bFGF) and transforming growth factor-beta 1 (TGF-β1), were present[5,17] in both bound and unbound fractions within the allograft.[17] This amnion/chorion tissue allograft was also reported to stimulate cell proliferation and promote mesenchymal stem cell (MSC) migration in vitro.[17] Collectively, these findings indicate that allografts from perinatal tissues, such as eUTA, can deliver a complex, donor-consistent matrix of structural and regulatory proteins with inherent potential to support connective tissue integrity, recruit progenitor cells, and modulate local inflammation.

Despite an established proteomic and cytokine profile,[15] critical questions remain regarding how the physical form of the eUTA scaffold and its associated vesicles govern the delivery of these proteins within injured musculoskeletal tissues. The micro- and nano-scale architecture of the microparticles will influence how cells adhere, infiltrate, and remodel the graft, while biodegradation characteristics will determine the timing and extent of protein release and the duration of mechanical support in the lesion environment.[13,16,18,19] In parallel, determining the presence of extracellular vesicles within the eUTA is needed to determine whether vesicle-borne cargo from this tissue would add additional paracrine signals that complement the matrix-bound and soluble proteins already identified.[5,20,21] By integrating microscopic, biodegradation, and extracellular vesicle data with the existing proteomic and cytokine profile, this study aims to move beyond cataloguing protein content to a more mechanistic understanding of how the eUTA behaves as a dynamic, bioactive scaffold in equine musculoskeletal repair.

Accordingly, the objective of this study was to expand the biological characterization of eUTA beyond bulk proteomics. The microscopic component aimed to (1) describe the persistence of the microstructure of fresh-thawed eUTA in a supportive tissue culture medium over two weeks and (2) evaluate two enzymatic digestion techniques to microscopically assess physical disaggregation of the microparticles in this tissue culture environment. The biological characterization component aimed to (1) assess the total degradation profile in saline or after targeted enzymatic digestion, (2) determine the presence and retention of key cytokines and growth factor mediators before and after detergent extraction, and (3) assess eUTA for the presence of extracellular vesicles and exosomes. We hypothesized that eUTA microparticles would naturally degrade over time and be susceptible to controlled enzymatic degradation, resulting in the progressive release of matrix and signaling proteins. This degradative profile, in combination with the defined microscopic architecture and associated extracellular vesicles, would support their function as a pro-reparative scaffold for equine musculoskeletal tissues.

Methods

Experimental Design

This study employed an in vitro, complementary microscopic and biochemical approach to characterize this micronized eUTA. For structural evaluation, eUTA, with and without enzymatic digestion, were assessed microscopically over time in a serum-containing tissue culture environment to determine the persistence of extracellular matrix architecture. Pellet-derived eUTA smears were assessed microscopically. To define degradation behavior, eUTA were incubated in saline or enzymatic solutions (collagenase I or trypsin) at physiological temperature, and temporal changes in particle morphology and protein release were quantified. The biological composition of eUTA was further characterized by measuring cytokine and growth factor profiles before and after Triton X-100 detergent extraction to assess the retention and extractability of key mediators. Finally, extracellular vesicles were isolated from eUTA and analyzed for concentration, size distribution, and expression of canonical surface markers. This design enabled an integrated assessment of the structural integrity, degradative dynamics, and associated bioactive components relevant to its proposed function as a pro-reparative extracellular matrix scaffold.

Equine Umbilical Tissue Allograft

The source of the eUTA evaluated in this study is available for investigative use and ethically sourced. Equicenta®CTM (Equine Performance Labs; Austin, TX) is a minimally manipulated umbilical-derived tissue allograft that is available as microparticles suspended in solution and cryopreserved. The umbilical-derived tissue allograft was obtained following full-term, natural delivery of vigorous foals, ensuring no harm to the mare or foal, and was sourced from a controlled Thoroughbred herd. The product was thawed at room temperature immediately before use in each portion of the study described below.

Experiment 1: EUTA Microscopic Evaluation in Biocompatible Medium and Conditions

Microscopic evaluation of eUTA in a serum-supported tissue culture medium over time was performed to determine whether the extracellular matrix architecture of eUTA persisted in a biocompatible tissue environment. Six culture flasks were allocated to receive no digestion, collagenase digestion, or trypsin digestion. For the no-digestion control, eUTA were fresh-thawed at room temperature and immediately placed into five mL of Dulbecco's Modified Eagle Medium (DMEM; Gibco; Grand Island, NY) with 10% fetal calf serum (FCS, Gibco; Grand Island, NY) in a T-25 flask (Corning; Tewksbury, MA) and incubated at 37°C and 5% CO_2. Fresh-thawed eUTA also underwent enzymatic digestion with either 50 U/mL Collagenase I (from Clostridium histolyticum; Millipore Sigma; Burlington, MA) or 0.25% trypsin (Gibco; Grand Island, NY) at a 1:1 ratio in 15 mL conical tubes and incubated at 37°C for one hour or 30 minutes, respectively. After incubation with enzymes, the tubes were centrifuged at 300xg for 10 minutes. The supernatant was discarded, and the pelleted eUTA product was resuspended, placed into separate T-25 flasks with five mls of DMEM with 10% FCS, and incubated at 37°C with 5% CO_2. Phase-contrast images were acquired at 10X and 20X magnification (EVOS® XL; Life Technologies; Carlsbad, CA) across multiple fields of view to document the appearance of the eUTA in the tissue supportive media. Biocompatible tissue cultures were maintained for up to two weeks with three mL of fresh media added on days six and ten.

Experiment 2: Preparation and Microscopic Evaluation of eUTA Pellet-Derived Smears

EUTA was thawed and pelleted via microcentrifugation at 10,000 rpm for five minutes. The supernatant was discarded, and the pellet was processed by direct smear to spread the concentrated tissue onto charged glass slides (Superfrost™Plus; Fisher Scientific, Waltham, MA, USA) using a method consistent with standard smear preparation techniques for particulate biological materials. Slides were fixed with a 1:1 dilution of cold acetone and methanol for 10minutes and then stained with hematoxylin (Electron Microscopy Sciences; Morgantown, PA, USA) for microscopic assessment (EVOS® XL; Life Technologies, Carlsbad, CA, USA) at 10x and 20x magnification.

Experiment 3: Microscopic and Biological Assessment of Degraded and Digested eUTA

A second enzymatic digestion experiment was performed using eUTAs from five different donors to assess the microscopic and biological effects of eUTA biodegradation. A vial of eUTA from each donor was equally divided into three groups at a 1:1 ratio with 1) saline (negative control), 2) collagenase I (100 U/mL; from Clostridium histolyticum; Millipore Sigma; Burlington, MA, USA) in saline, or 3) 0.25% trypsin (Gibco; Grand Island, NY, USA) and placed into their respective wells on a 24-well tissue culture plate (Corning, Tewksbury, MA, USA). Separate wells were used for each time point evaluated (30 minutes (one-half hour), two hours, and 24 hours), and plates were incubated at 37°C and 5% CO_2 for the duration of the study period. Microscopic images were captured from each well for 30 minutes (one-half hour), two hours, and 24 hours using a phase microscope (EVOS® XL; Life Technologies; Carlsbad, CA, USA) at 10X magnification. After microscopic evaluation, eUTA samples from each well were collected at their respective time points and frozen at −80°C until protein analysis and ELISAs were performed.

The total protein concentration for each sample was determined using Bradford analysis. Briefly, 250 µL of Quick Start Bradford 1X dye was added to each well of a 96-well plate, along with five µL of undiluted supernatant from each donor's eUTA in saline or from eUTA digested in collagenase or trypsin. Serial dilutions of bovine serum albumin (BSA)

included in the Bradford kit (Bio-Rad, Hercules, CA, USA) were used to generate a standard curve. Saline was placed in blank wells, and all samples and standards were analyzed in duplicate. Saline, collagenase, and trypsin without eUTA at a 1:1 dilution with saline were evaluated in duplicate as assay controls. After five minutes of incubation at room temperature, the 96-well plates were analyzed using a Synergy Biotek Spectrophotometer (Winooski, VT, USA) at 595 nm. Concentrations of Tissue Inhibitor of Metalloproteinases-2 (TIMP-2) and collagen type 1 alpha chain (COL1A1) were determined using ELISA (RayBiotech Life, Inc.; Peachtree Corners, GA, USA) from the supernatants.

Experiment 4: Biological Mediator Assessment from Detergent-Extracted eUTA

Concentrations of selected pro- and anti-inflammatory mediators, growth factors, and extracellular protease inhibitors were measured in a separate set of 10 eUTA samples, each derived from a different umbilical cord donor. Samples were analyzed before and after extraction with Triton X-100 (Acros Organics, Thermo Fisher Scientific, Waltham, MA, USA), with both extraction and multiplex ELISA testing performed by RayBiotech (Peachtree Corners, GA, USA). Concentrations were determined using an equine-specific multiplex ELISA kit (RayBiotech Life, Inc., Peachtree Corners, GA, USA). Samples were run in duplicate and compared with standard curves to quantify each analyte as previously described.[15] Mediators assessed included anti-inflammatory mediators (interleukin-1 receptor antagonist (IL-1ra), interleukin-4 (IL-4), and interleukin10 (IL-10)), pro-inflammatory mediators (interferon gamma (IFNγ), interleukin-1 alpha (IL-1α), interleukin-2 (IL-2), interleukin-8 (IL-8), interleukin-15 (IL-15), and monocyte chemoattractant protein-1 (MCP-1)), anabolic mediators (insulin growth factor-1 (IGF-1), basic fibroblastic growth factor (bFGF), transforming growth factor-beta-1 (TGF-β1), and vascular endothelial derived growth factor (VEGF)), and anti-catabolic mediators (alpha-2-macroglobulin (A2M), tissue inhibitor of metalloproteinases-1 and −2 (TIMP-1 and TIMP-2)). Mediators were categorized as anti-inflammatory, pro-inflammatory, anabolic, or anti-catabolic based on their function, and summary indices were calculated before and after Triton X-100 extraction.

Experiment 5: EUTA Extracellular Vesicle (EV) Isolation and Characterization

Equine UTA from 11 individual donors, as previously described[15] were used to isolate and characterize EVs (Creative Proteomics; Shirley, NY, USA). Briefly, the eUTAs were microcentrifuged at 2,000 × g for 10 minutes at 4°C and the supernatant was transferred to a 22.5 mL thick-walled polycarbonate tube (Beckman Coulter, Indianapolis, IN, USA). Each sample was then ultracentrifuged (Optima L-90K; Beckman Coulter, Indianapolis, IN, USA) at 35,600 rpm for 90 minutes at 4°C. The resultant pellet from each sample was resuspended in 800 µL of filtered phosphate-buffered saline (PBS). The samples were ultracentrifuged (Optima TLX-120; Beckman Coulter, Indianapolis, IN) at 65,000 rpm for 90 minutes at 4°C. The final pellet was resuspended in 100 µL of PBS for downstream characterization and analysis.

The isolated EVs were analyzed using the NanoView ExoView®platform (NanoView Biosciences; Brighton, MA, USA). The samples were diluted according to the manufacturer's guidelines to achieve optimal particle loading on ExoView chips pre-coated with antibodies against CD9, CD63, and CD81 for particle capture. Diluted samples were incubated on chips overnight at room temperature to allow for exosome binding. After incubation, the chips were washed and fluorescently labeled detection antibodies (CD9, CD81, CD63, CD90, and CD105) were applied to identify the markers carried by each EV. After additional washes to remove excess detection antibodies, the chips were imaged using an ExoView R100 reader. Interferometric imaging provided single-particle sizing and overall EV counts, and fluorescent imaging enabled marker-specific profiling and co-localization analysis of individual particles. The total captured particle counts, size distribution of captured particles, and presence of the relative abundance of tetraspanins (CD9, CD63, and CD81), CD105, and CD90 to confirm isolated EVs as exosomes were determined using the ExoView Analyzer software. In accordance with the International Society for Extracellular Vesicles (ISEV), not only was the eUTA assessed for the presence of transmembrane EV markers (ie, tetraspanins) but also for the presence of a cytosolic marker (ie, ALIX/PDCD61P or TSG101) to demonstrate that the vesicles contain enclosed intracellular material; therefore, subsequent proteomic analysis via mass spectrometry of the EVs was performed.

Data Analysis

Microscopic images were qualitatively assessed, with descriptions of the findings included in the results. Total protein, COL1A1, and TIMP-2 concentrations after enzymatic digestion were analyzed using a mixed-effects model (JMP®19; JMP Statistical Discovery LLC; Cary, NC, USA). Treatment, time, and their interactions were included as fixed effects and donors were included as random effects. Least-squares means were compared using Tukey's post-hoc test. Model assumptions were assessed by examining the residual plots. The normality of residuals was evaluated using quantile-quantile (QQ) plots. Due to deviations from normality, the COL1A1 data were log-transformed prior to analysis. Pre- and post-extraction data were analyzed using paired t-tests for each assessed cytokine and growth factor. The mean number and size of isolated EVs per donor vial were reported, along with a description of the EV markers assessed. The results were considered statistically significant at $p<0.05$.

Results

eUTA Microscopic Assessment in Biocompatible Medium and Pellet-Derived Smears

Phase-contrast microscopy revealed an intact, scaffold-like structure of eUTA suspended in the biocompatible tissue culture medium supplemented with FCS, consistent with the behavior expected of extracellular matrix-derived biomaterials under physiological conditions. The eUTA that was not enzymatically digested formed large extracellular matrix aggregates that did not adhere to the bottom of the plastic flask and instead floated in the culture medium (Figure 1, no digestion). Although similar in appearance and behavior, one hour of enzymatic digestion with collagenase prior to placement in the tissue culture medium supplemented with FCS resulted in smaller and fewer matrix aggregates, indicating partial collagen digestion and reduced particulate conformation (Figure 1, collagenase digestion). Thirty minutes of trypsin exposure prior to placement in the biocompatible medium resulted in almost complete digestion of the eUTA, and only a few particle aggregates remained (Figure 1, trypsin digestion). The eUTA matrix obtained from both collagenase and trypsin digestion remained suspended in the medium and did not adhere to the flask. No further breakdown or degradation of eUTA aggregates was observed over the following weeks in biocompatible tissue culture media containing 10% FCS while maintained at 37°C. Brightfield microscopy of the pelleted eUTA smear stained with cytological techniques revealed a fibrillar, filamentous extracellular matrix with a distinct net-like architecture that complemented the 3-D structure (Figure 2).

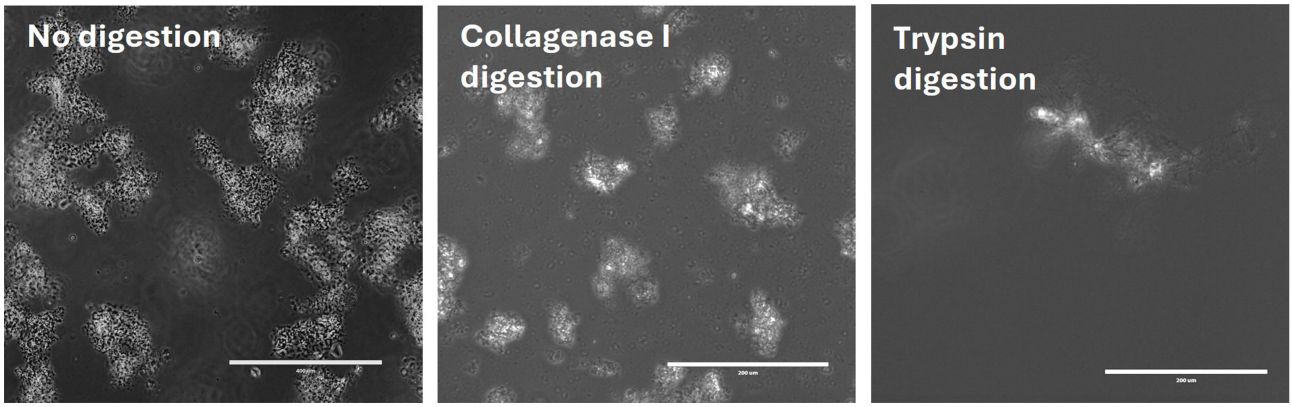


Figure 1 Microscopic qualitative evaluation of eUTA in a biocompatible tissue culture medium alone (no digestion), after one hour of collagenase I digestion and after 30 minutes of trypsin digestion. In culture media, the eUTA contained aggregates of extracellular matrix that did not adhere to the plastic and floated within the fluid, as would be expected for matrix particles. After enzymatic digestion with collagenase (middle image), the aggregates appeared smaller, indicating collagen digestion and partial loss of particulate form. Trypsin digestion of particles (right image) was rapid and near complete compared to the control (no digestion; eUTA in media) and collagenase digested eUTA cultures as represented by the images. Particulate size changed immediately after digestion, and eUTA microparticles maintained their appearance for the rest of the experiment. No digestion Bar=400 um; Collagenase I and trypsin Bar=200 um.

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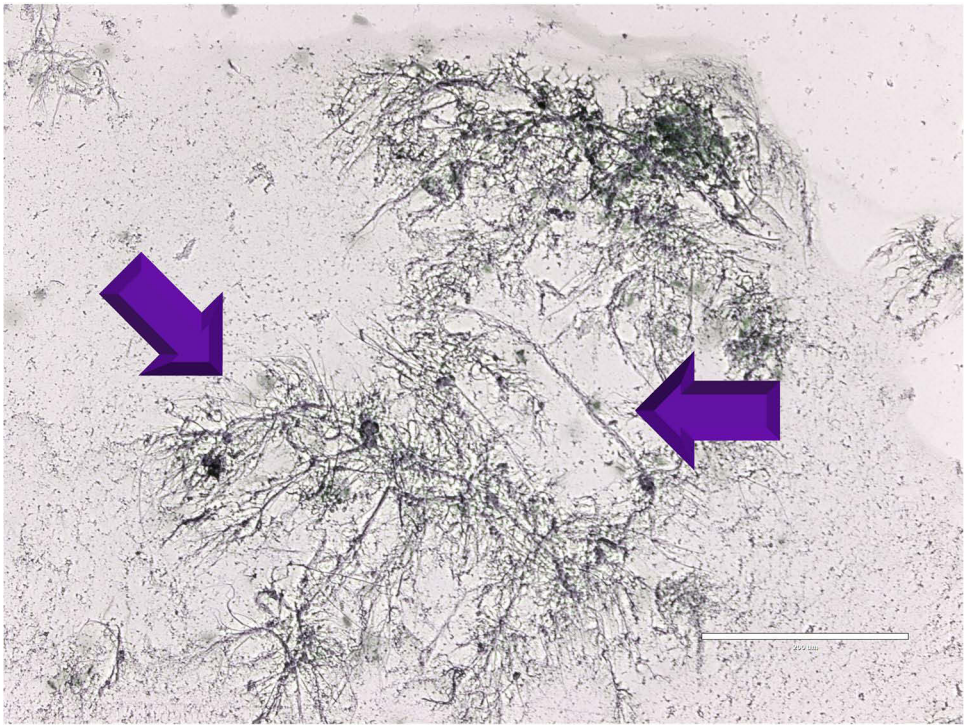


Figure 2 Cytological evaluation of pelleted umbilical tissue allograft fixed with cold methanol:acetone and stained with hematoxylin. Note the presence of filamentous-form extracellular matrix (purple arrows) creating a net-like protein scaffold structure. Bar =200 um.

eUTA Biologic Degradation and Enzymatic Digestion

Exposure of eUTA to saline, collagenase, or trypsin in saline at 37°C resulted in a reduction in the number, size, and networking of filaments and total protein over time (Figure 3). eUTA showed the least particle degradation in saline tissue wells compared to enzymatically digested eUTA. Correspondingly, there was a significant decrease ($p<0.001$) in total protein concentration over time in all groups (Figure 3A). For specific proteins, collagenase digestion rapidly solubilized COL1A1 in the eUTA, with significant increases in COL1A1 concentration of 197.7±174.6 at two hours and 750.3±673.2 at 24 hours of incubation at 37°C compared with barely detectable concentrations with saline or trypsin ($p<0.001$; Figure 3B). There was a significantly greater ($p<0.05$) concentration of TIMP-2 in eUTA with collagenase (1078±158.4 ng/mL) and trypsin (1476.6±158.4 ng/mL) digestion compared with saline alone (443±158.4 ng/mL) at one-half hour of incubation at 37°C and a further significant increase in TIMP-2 after collagenase (1133.8±158.4 ng/mL) and trypsin (1135.9±158.4 ng/mL) digestion compared with saline (606.8±158.4 ng/mL) at two hours. TIMP-2 was not detectable after 24 hours of collagenase or trypsin exposure (Figure 3C).

eUTA Biological Mediators After Detergent Extraction

Overall, the anti-catabolic ($p<0.001$) and anabolic ($p=0.025$) indices were significantly greater after extraction than before extraction (Figure 4). Specifically, alpha-2-macroglobulin (A2M; $p=0.004$) and tissue inhibitor of metalloproteinases-2 (TIMP-2; $p=0.018$) were increased post-extraction by 2.7-fold and 3.5-fold, respectively. Anabolic growth factors, specifically insulin-like growth factor-1 (IGF-1; $p=0.025$), increased post-extraction 2.3-fold. Anti-inflammatory biomediators, as well as the index, were modestly low in eUTA and were lower after matrix extraction, indicating that these cytokines are likely soluble, rather than matrix bound. ($p=0.019$). IL-1Ra, the receptor antagonist for the inflammatory cytokine Il-1, had the greatest pre-extraction concentration (269.9±0.18 pg/mL, $p=0.017$) among the anti-inflammatory cytokines evaluated (Figure 4). Pro-inflammatory cytokines were low in eUTA and decreased post-extraction ($p=0.028$; Figure 4), indicating eUTA does not harbor inflammatory cytokines.

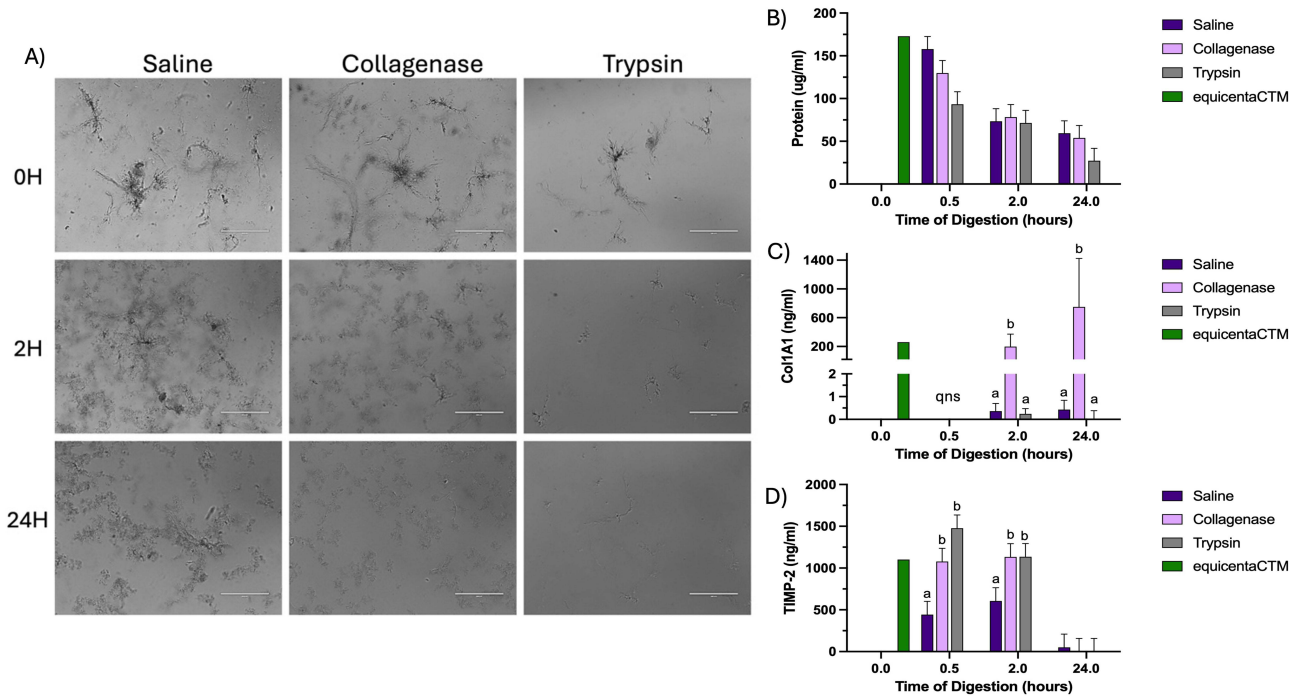


Figure 3 Microscopic and protein analysis of the eUTA at 1:1 ratio with saline or with collagenase or trypsin in saline (Bar = 400 um). (**A**) Microscopic images showed a loss of the net-like matrix structure over time (0H, 2H, and 24H) under all conditions (saline, collagenase, and trypsin), with fewer filaments observed after two and 24 hours, indicating degradation when maintained at 37°C. Qualitative assessment showed greater loss of matrix structure with collagenase and trypsin digestion compared with the saline control. (**B**) Graph B (top right) demonstrates least squares means ± SEM for protein concentration (ug/mL) of eUTA after one-half, two and 24 hours in saline or digestion with collagenase or trypsin in saline. Note that total protein concentration of the eUTA decreased over time ($p<0.001$) for all conditions. (**C**) Graph C (middle right) demonstrates COL1A1 concentrations (ng/mL) of eUTA after one-half, two and 24 hours in saline or digestion with collagenase or trypsin in saline. Data are presented as back-transformed least-squares means ± SEM, as statistical analyses were performed on log-transformed data. Different letters (a, b) indicate significant differences between treatments ($p < 0.001$; qns = quantity not sufficient for one-half hour). There was a significantly greater concentration of COL1A1 in eUTA digested with collagenase than in saline or trypsin. (**D**) Graph D (bottom right) demonstrates the least squares means ± SEM for TIMP-2 concentrations (ng/mL) of eUTA after one-half, two and 24 hours in saline or digestion with collagenase or trypsin in saline. Different letters (a, b) indicate significant differences among treatments ($p < 0.05$) at each time point. Note that eUTA in saline had significantly less TIMP-2 at one-half and two hours compared to the eUTA digested with collagenase or trypsin, suggesting TIMP-2 release as the eUTA matrix was broken down. Immediately thawed, undigested eUTA run at the same time is included on each graph for reference.

eUTA Extracellular Vesicle (EV) Isolation and Characterization

Extracellular vesicle analysis confirmed a mean ± SEM of $9.28\times10^{11} \pm 2.2\times10^{10}$ vesicles/mL with a mean ± SEM particle size of 133.98±0.7 nm (Figure 5). The particle count for the CD9 capture probe was 10^8–10^9 particles/mL, which far exceeded that of the isotype control. CD81 was also detected in abundance by the capture probe. Overall, 63% of the particles co-localized with the CD9 fluorescent antibody, 11.6% with the CD81 fluorescent antibody, and 22.1% with both CD9 and C81 (Figure 5 and Table 1). Although the direct fluorescent antibody for CD63 was not specific for equine, the CD63 capture probe on the ExoView chip effectively bound vesicles from the samples above the isotype control. CD63-captured EVs also co-localized with fluorescent antibodies against CD9 and CD81. The mesenchymal cell markers CD90 and CD105 co-localized with the capture probes CD9, CD81, and CD63 on the ExoView chip in a small percentage (<5%) of captured vesicles. Proteomic analysis of the EV samples revealed a high abundance of the cytosolic protein ALIX (ALG-2 Interacting Protein X).

Discussion

This study is the first to demonstrate that this eUTA is composed of a biologic, filamentous net-like scaffold containing soluble and extracellular matrix proteins with the potential to serve in vivo as a structural framework for endogenous cell migration and time-dependent release of pro-reparative proteins. The growth- and development-associated proteins identified in this study are known to facilitate cell recruitment and support the local tissue environment during healing. Microscopic assessment of eUTA confirmed micron-sized particles that tended to aggregate in close proximity, and

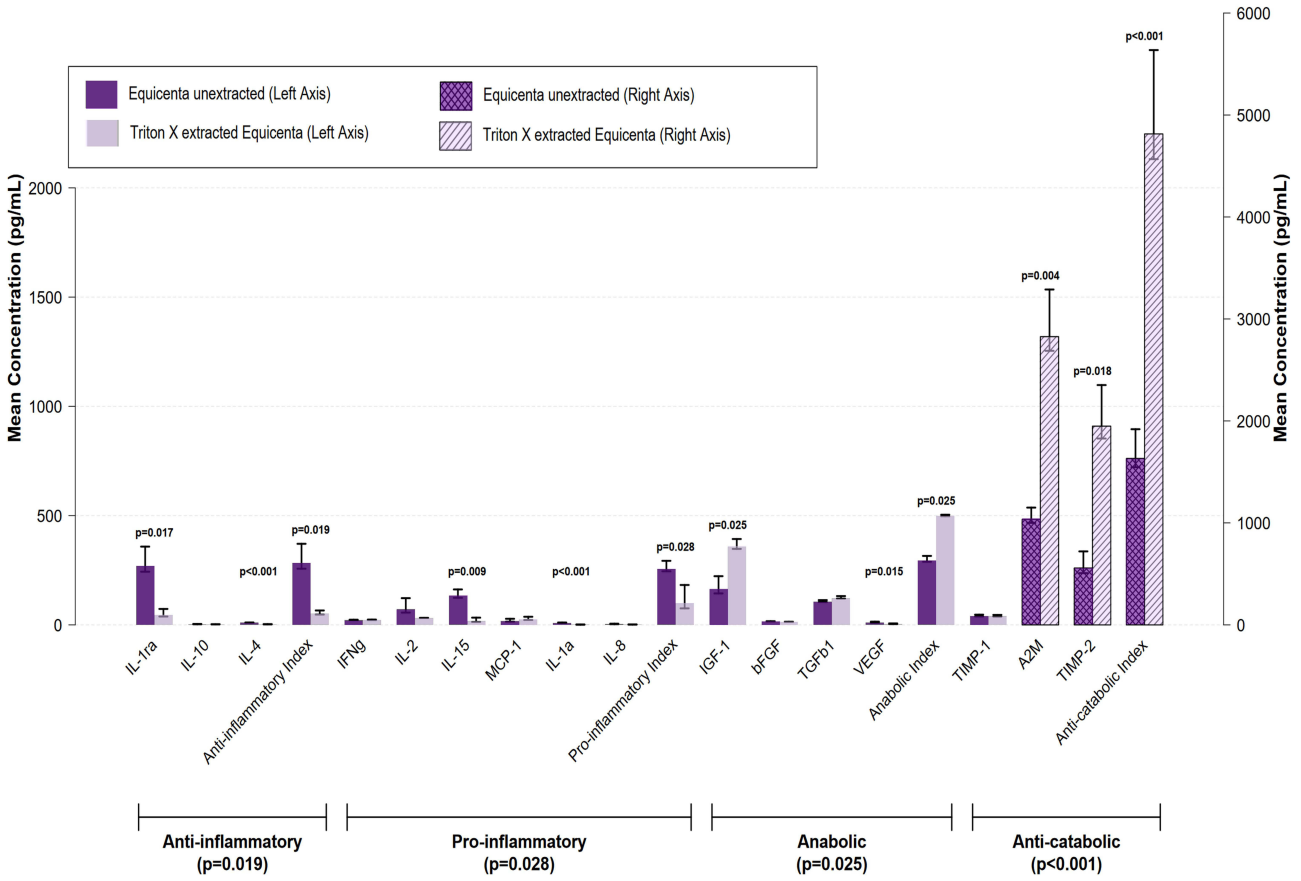


Figure 4 Cytokine and growth factor concentrations of the eUTA before (dark purple) and after (light purple) Triton X-100 extraction. Note the significant decrease in inflammatory indices (anti-inflammatory, p=0.019 and pro-inflammatory, p=0.028) and significant increase in anabolic (p=0.025) and anti-catabolic (p<0.001) indices, with specifically marked increases in IGF-1, A2M, and TIMP-2 post-extraction.

phase-contrast imaging revealed gelatinous matrix-like clusters similar to those reported for other commercially available minimally manipulated Wharton's jelly allografts.[13]

Bacterial collagenase partially degraded the eUTA bioscaffold, resulting in reduced clumping and smaller particle aggregates. These observations are consistent with degradation of the collagen-rich Wharton's jelly matrix from which eUTA is derived.[13] Although bacterial collagenase differs from endogenous mammalian collagenases in substrate specificity and enzymatic activity, the results demonstrate that the matrix is vulnerable to collagenolytic degradation. Therefore, it is plausible that endogenous proteolytic enzymes present within injured tissues could contribute to the natural remodeling and biodegradation of eUTA in vivo. Digestion with 0.25% trypsin produced the lowest amount of aggregation, consistent with more extensive degradation of the bioscaffold matrix than observed following collagenase treatment. This result was anticipated because trypsin was included as a positive control to demonstrate susceptibility of the matrix to enzymatic digestion. In addition to the microscopic findings, collagenase and trypsin digestion for 24 hours in saline reduced the total protein concentration, further indicating that eUTA proteins can be enzymatically degraded. Collagenase specifically promoted the release and solubilization of matrix-bound collagen fragments, as evidenced by higher COL1A1 concentrations, which continued to increase over time. Saline alone at 37°C also caused degradation, likely through hydrolysis, mechanical weakening of the matrix, and intrinsic enzymatic activity, particularly in the unbuffered acidic non-tissue-supportive saline medium.

Exposure to collagenase and trypsin for 30 minutes and two hours, increased TIMP-2 concentrations, indicating release of this factor from the eUTA matrix. Similarly, biochemical extraction with Triton X-100 resulted in increased concentrations of IGF-1, TGF-β, TIMP-1, and A2M. The elevated levels of these anabolic and anti-catabolic factors are

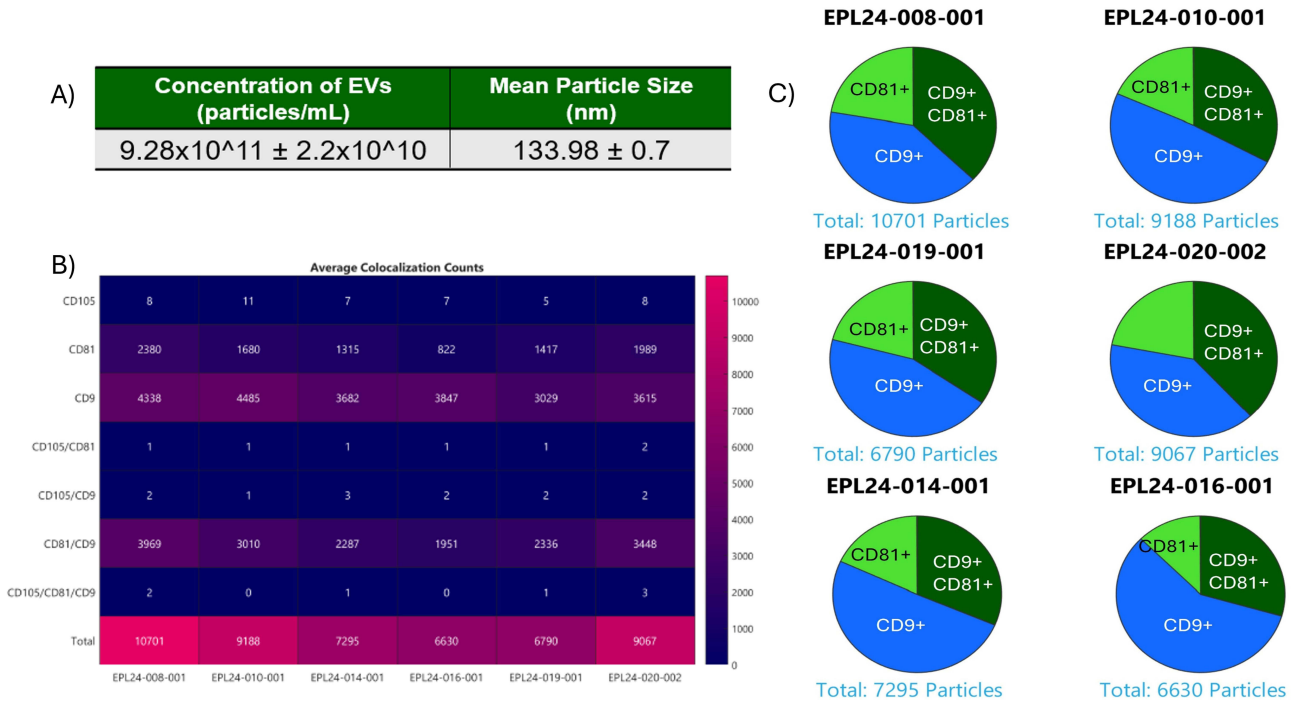


Figure 5 Extracellular vesicle (EVs) determination in eUTA. (**A**) Size and marker analyses confirmed that hundreds of billions to trillions of EVs were present in eUTA. (**B**) Demonstration of average co-localization counts from the nanoparticle tracking analysis. CD9 capture probe on the ExoView Chip followed by fluorescent antibody co-localization of the tetraspanins CD9 and CD81 and mesenchymal stem cell (MSC) marker CD105. Average counts for co-localization with CD9 and with CD9 and CD81 were the highest. (**C**) Pie charts showing ratios of assessed tetraspanin markers.

consistent with previous reports identifying Wharton's jelly as a reservoir of matrix-bound bioactive proteins.[22] While Triton X-100 is a nonphysiological detergent and does not directly mimic a specific endogenous mediator, tissue injury is associated with the production of proteases (eg, matrix metalloproteinases, plasmin, and neutrophil-derived serine proteases), phospholipases, and reactive oxygen species that collectively contribute to extracellular matrix remodeling, cellular membrane disruption, and release of matrix-associated proteins. Accordingly, Triton X-100 was used as a biochemical extraction tool to determine whether eUTA contains matrix-associated bioactive factors that could potentially become available during tissue remodeling, rather than model a specific in vivo degradation mechanism.

Table 1 Colocalization Percentage (±SD) of the Fluorescent Antibodies for Each Tetraspanin Capture Probe

Fluorescent antibody	Capture Probe						
	CD9		CD81		CD63		AVG
	%	±SD	%	±SD	%	±SD	%
CD90	0.2	0.2	0.2	0.2	6.8	3.7	2.4
CD81	14.3	6.5	8.7	7.5	11.8	4.6	11.6
CD9	55.2	13.1	65.9	13.0	68.0	10.4	63.0
CD90/CD81	0.0	0.0	0.0	0.0	0.3	0.6	0.1
CD90/CD9	0.2	0.4	0.2	0.5	1.0	1.3	0.5
CD81/CD9	29.9	7.8	24.7	6.6	11.7	5.9	22.1
CD90/CD81/CD9	0.2	0.3	0.2	0.4	1.3	1.6	0.5

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Wharton's jelly is a well-recognized source of peptide growth factors and, despite its lower cellularity than umbilical vasculature, contains higher concentrations of growth factors per microgram of DNA.[22] This study confirmed that eUTA similarly functions as a reservoir of protein growth factors and biomediators that can be released over time. Growth factors in Wharton's jelly are believed to be tightly associated with high–molecular weight extracellular matrix components,[22] and several factors in this study increased after detergent extraction, supporting release from matrix-bound stores. The cytokine and growth factor profiles of eUTA, including TIMP-1, TIMP-2, bFGF, VEGF, and IGF-1, are comparable to other human perinatal-derived ECM products, some of which have been shown in vivo to enhance angiogenesis and reduce fibrosis.[18] VEGF concentrations were relatively low but similar to those reported in human amnion/chorion allografts that nevertheless promote endothelial proliferation and angiogenic signaling.[19]

Among the anabolic mediators, bFGF, a chondrocyte mitogen that upregulates cartilage repair and matrix synthesis pathways,[23] was readily detectable, with higher concentrations before extraction and levels comparable to or exceeding those reported for adult autologous biologics such as PRP and bone marrow aspirate.[14] IL-1Ra, an anti-inflammatory antagonist of IL-1 signaling, was present at lower concentrations than those reported for autologous conditioned serum,[24,25] likely reflecting differences in tissue source and processing; however, in vivo eUTA injection into equine articular tissues elicited a marked increase in IL-1Ra and a sustained elevation of the IL-1Ra:IL-1 ratio for over 3 weeks.[15] Triton X-100 extraction, by disrupting lipid-protein interactions and solubilizing matrix-bound proteins, further increased measurable anabolic and anti-catabolic mediators, and enzymatic digestion with collagenase or trypsin similarly facilitated the release of matrix-associated proteins. Collectively, these findings indicate that eUTA serve as a reservoir of bioactive factors that can be progressively liberated as the matrix degrades, potentially enhancing the bioavailability of key regulatory molecules in the local microenvironment.

ExoView analysis of eUTA from multiple donors identified hundreds of billions to trillions of extracellular vesicles (EVs) in the 130–134 nm range, consistent with small EVs, including exosomes.[26,27] These vesicles expressed canonical exosomal markers (CD9, CD63, and CD81) and contained the cytosolic protein ALIX, confirming their exosomal identity, and a subset expressed MSC-associated markers such as CD90 and/or CD105. These data support the presence of matrix-bound nanovesicles within the eUTA retained after processing. EVs and matrix-bound nanovesicles[20,28] are increasingly recognized as mediators of tissue repair through delivery of microRNAs and other regulatory cargo that modulate macrophage polarization,[20] extracellular matrix remodeling, and structural regeneration.[29] The presence of EVs within the eUTA suggests an additional mechanism by which this allograft may contribute to tissue healing, functioning not only as a structural scaffold but also as a reservoir of vesicle-borne biological signals.

The limitations of this study include a subset of qualitative microscopic assessments and an in vitro model that cannot fully recapitulate the complexity of the in vivo equine environment, or the spectrum of proteolytic processes present in injured tissues. Nonetheless, clear and consistent differences were observed between undigested and digested products, and the combined cytokine, growth factor, and EV data provide a coherent picture of eUTA as a biologically active extracellular matrix scaffold. Through their native composition and biodegradation profile, eUTA appear to be capable of providing structural support while enabling the progressive release of growth factors, anti-catabolic mediators, and EVs, which may collectively influence cellular recruitment, matrix remodeling, inflammatory regulation, and ultimately functional repair in equine musculoskeletal injuries.

Conclusion

In conclusion, the eUTA evaluated in this study, exhibited key biomaterial properties by supplying a biological extracellular matrix that may provide structural support and serve as a reservoir of signaling molecules that may promote tissue healing in orthopedic injuries. These characteristics may enhance endogenous repair processes and improve the outcomes of equine musculoskeletal injuries.

AI Assistance Statement

Grammarly (2026) and Microsoft Copilot (Microsoft, 2026) were used to improve the language, grammar, and overall readability of the manuscript. No AI tools were used to generate scientific content, analyze data, or influence the

conclusions of this study. All content was reviewed and approved by the authors. Additionally, the authors checked the terms of use for the specific AI tool used and confirmed its suitability for publication; they take full responsibility for the integrity of the content, including the accuracy of references.

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Author Contributions

All authors made a significant contribution to the work reported, whether that is in the conception, study design, execution, acquisition of data, analysis and interpretation, or in all these areas; took part in drafting, revising or critically reviewing the article; gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work.

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